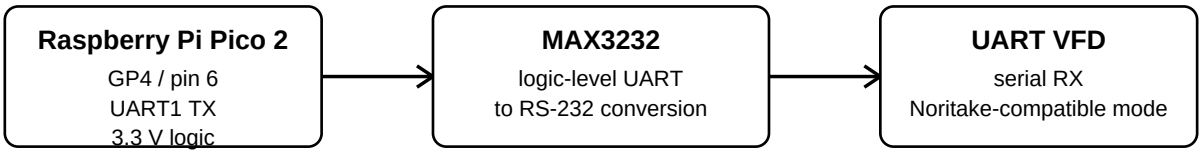
AAC Consumer Edition - Optional UART VFD Interface

NORITAKE-COMPATIBLE CHARACTER VFD | REVISION A | INTERNAL / PRIVATE - DIMENSION EIGHT LABS

Device / Point	Pin / Interface	Connects To	Purpose / Notes
Raspberry Pi Pico 2	GP4 / physical pin 6	MAX3232 logic-side TX path	UART1 transmit only. Optional VFD driver owns GP4 only when selected.
Pico 2 UART	UART1 / 9600 baud / 8N1	MAX3232 level converter	3.3 V logic UART converted to RS-232 signaling for the display.
UART RX	Not used	-	VFD path is TX-only. No Pico RX GPIO is allocated; GP5 remains free.
MAX3232	RS-232 output	VFD serial RX input	Use the actual transmit signal path, not assumptions based only on module silkscreen.
Ground	Common reference	Pico / MAX3232 / VFD signal ground	All serial-interface grounds must share a common reference.
Display power	Per display specification	VFD power input	Do not infer VFD power requirements from the UART interface.

Signal Path



Example: Posiflex PD-2200 used by the original AAC. TX-only; no return serial path to the Pico.

Noritake-Compatible Driver Notes

Item	Behavior / Guidance
Display mode	Configure the VFD for its Noritake-compatible serial command mode before using this driver.
Character output	The PD-2200 backend sends printable character data plus Noritake-compatible control / positioning commands. Its renderer is built around a 20-column x 2-row character display.
Clear command	`0x0E` was experimentally verified on the AAC Posiflex PD-2200 to clear displayed characters **without resetting the cursor / write position**.
Protocol authority	For exact command bytes and device quirks, use the original AAC / current VFD driver source as the implementation reference. Preserve experimentally verified PD-2200 behavior separately from behavior inferred from later Posiflex manuals.

Bench Wiring Notes - Read Before Connecting

• **Optional display:** This interface is used only when the UART VFD driver is selected.

• **TX only:** GP4 is the only Pico header GPIO required by the VFD driver.

• **MAX3232 module warning:** The Chinese module used during AAC development had misleading TX/RX labeling and nonstandard DB9 wiring. Verify the actual signal path instead of trusting the silkscreen or connector convention.

• **Do not bypass level conversion:** The Pico UART is logic-level; a true RS-232 VFD input requires the MAX3232 (or equivalent) stage.

• **Grounding:** Pico, level converter, and display serial ground must be common.

• **Display configuration:** Confirm the VFD is in Noritake-compatible mode and set for 9600 baud, 8 data bits, no parity, 1 stop bit.

DO NOT CONNECT A TRUE RS-232 SIGNAL DIRECTLY TO A PICO GPIO.

Status: Reference sheet for the optional UART/Noritake-compatible VFD path. The original AAC remains the primary working reference for the Posiflex PD-2200 implementation.